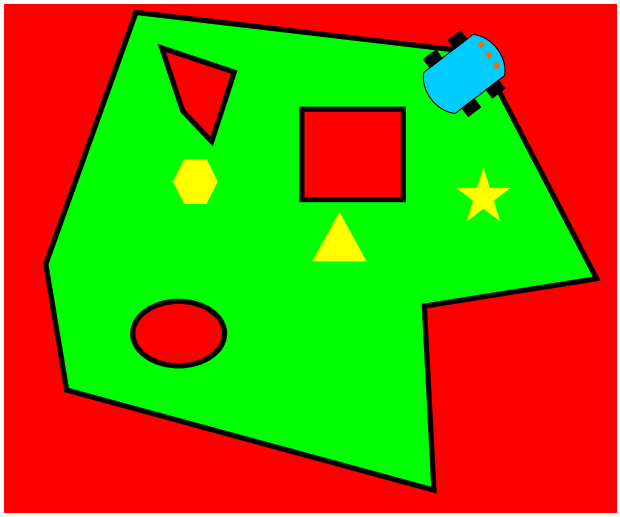
Toronto Metropolitan University

CPS 607 – Autonomous Mobile Robotics
Lab 2 – Object avoidance

The Environment and Concepts:

There exists a flat world of (possibly) irregular shape. The world is surrounded by a void. Somewhere in the test environment, there exists objects placed in unknown configurations that need to be avoided.

Required:



- Working in groups of 1 to 2, you are to create an Autonomous Mobile Robot (AMR) capable of surviving in the world while in (near) continuous motion for a period of no less than 2 minutes without falling off the "edge."
- The robot must also avoid collision with any objects (as described above) on the surface (NOTE: the robot may touch the object but have to show signs of detecting it and change direction to avoid it.)
- The formal definition of near continuous motion is that the robot should always be moving, and the robot can stop moving for no longer than 1 second.
- Students may not know what the objects will be ahead of time or where they will be placed on the surface. Anticipating what type of objects may exist is part of the lab.
- Objects can be any shape, will weigh at least 100g, can be made of any type of material. Objects are solid. Object's size will be at least 30 mm^3. Objects may not be stationary but will not approach the robot from behind.
- You must make a short video (less than 2 minutes) of your robot on the specified environment.
- You must create a writeup (1 page of text maximum) describing of your algorithm/strategy and cover these points: Describe the overall design of the robot. Describe what sensors you used, why they were chosen, and the placement of the sensors and why that is important.

Submissions:

- You must demo your robot at the arranged time slot.
- You must submit your video on D2L on the day of demo.
- You must submit the 1-page writeup on D2L on the day of demo. (NOTE: writeups are not accepted if the robot is not demoed)

Late submission is possible. Since this has to be demoed, the only other time to submit would be the following week. You'll get a 20% deduction for late submission. Anything later than 1 week will not be accepted.

Rubrics:

The lab will be marked out of 40. All marks will come from the demo, but no marks will be given without submission of the writeup.